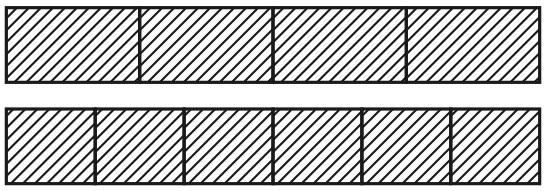
Fractions and Wholes

All the pieces make 1 whole. More pieces go past 1.

All the pieces: 1 whole

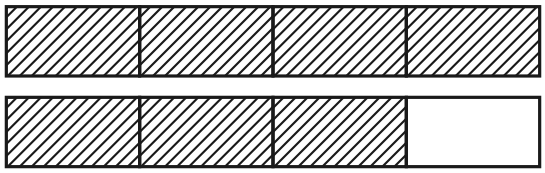


$\frac{4}{4} = 1$

$\frac{6}{6} = 1$

Top and bottom the same? Then it's 1 whole.

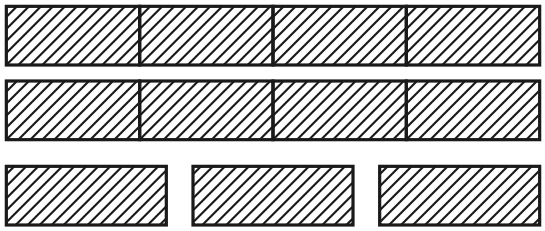
More than 1 whole



$\frac{7}{4} = 1 \text{ whole and } \frac{3}{4}$

Count on past 4/4: 5/4, 6/4, 7/4.

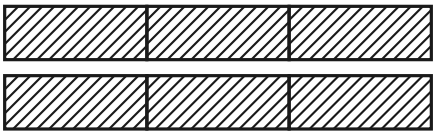
Whole numbers are fractions too



$\frac{8}{4} = 2$ 8 fourths fill 2 bars.

$\frac{3}{1} = 3$ 3 pieces, each a whole.

Your turn!



$\frac{6}{3} = 2$

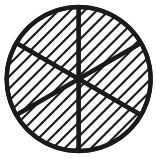
For teachers

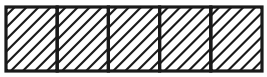
Goal: b/b = 1, fractions bigger than 1, and whole numbers as fractions (3.NF.3c). Say “How many pieces make one whole bar? Count on past it.” No mixed-number rules yet: 7/4 is 1 whole and 3/4.

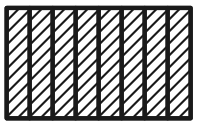
Fractions and Wholes: Practice

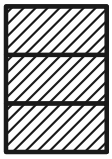
When the top number matches the bottom, that's 1 whole.

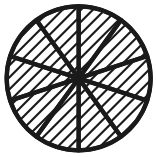
Part 1 One whole Every part is shaded. How many pieces?

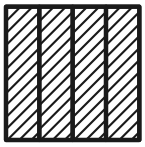
1)  $\frac{6}{6} = 1$

2)  $\frac{5}{5} = 1$

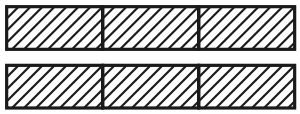
3)  $\frac{8}{8} = 1$

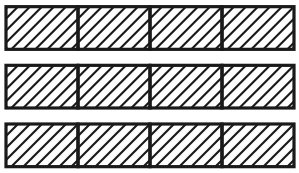
4)  $\frac{3}{3} = 1$

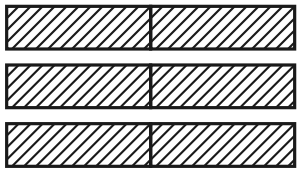
5)  $\frac{10}{10} = 1$

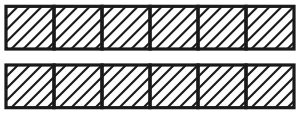
6)  $\frac{4}{4} = 1$

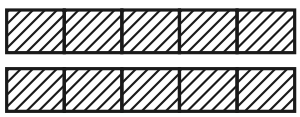
Part 2 How many wholes? Count the pieces, then the whole bars.

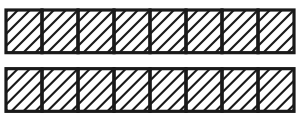
7)  $\frac{6}{3} = 2$

8)  $\frac{12}{4} = 3$

9)  $\frac{6}{2} = 3$

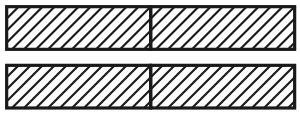
10)  $\frac{12}{6} = 2$

11)  $\frac{10}{5} = 2$

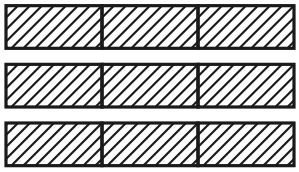
12)  $\frac{16}{8} = 2$

Fractions and Wholes: Practice (continued)

Part 2 How many wholes? (continued)

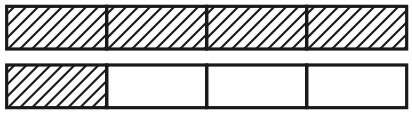
13) 

$\frac{4}{2} = 2$

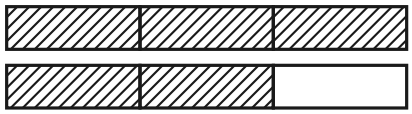
14) 

$\frac{9}{3} = 3$

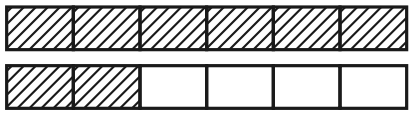
Part 3 More than one whole Full bars first, then the pieces left.

15) 

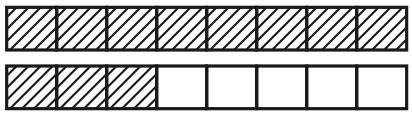
$\frac{5}{4} = 1 \text{ whole and } \frac{1}{4}$

16) 

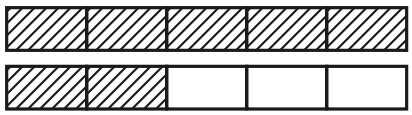
$\frac{5}{3} = 1 \text{ whole and } \frac{2}{3}$

17) 

$\frac{8}{6} = 1 \text{ whole and } \frac{2}{6}$

18) 

$\frac{11}{8} = 1 \text{ whole and } \frac{3}{8}$

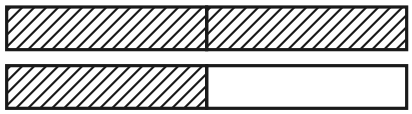
19) 

$\frac{7}{5} = 1 \text{ whole and } \frac{2}{5}$

Fractions and Wholes: Practice (continued)

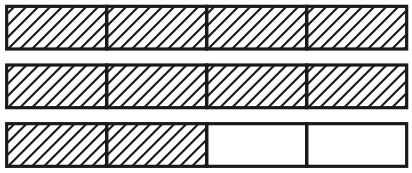
Part 3 More than one whole (continued)

20)



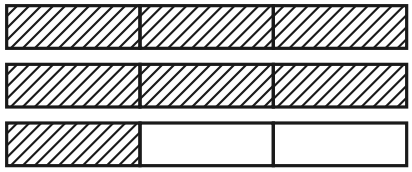
$\frac{3}{2}$ = **1 whole and** $\frac{1}{2}$

21)



$\frac{10}{4}$ = **2 wholes and** $\frac{2}{4}$

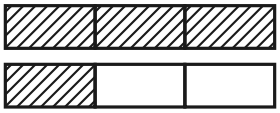
22)



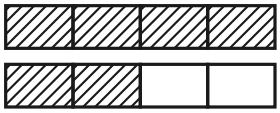
$\frac{7}{3}$ = **2 wholes and** $\frac{1}{3}$

Part 4 Shade it

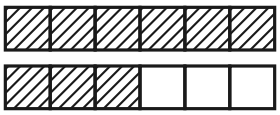
23) **Shade** $\frac{4}{3}$



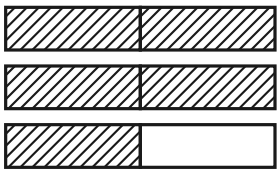
24) **Shade** $\frac{6}{4}$



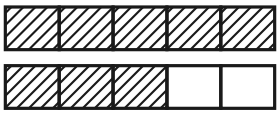
25) **Shade** $\frac{9}{6}$



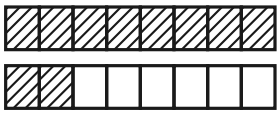
26) **Shade** $\frac{5}{2}$



27) **Shade** $\frac{8}{5}$



28) **Shade** $\frac{10}{8}$



Fractions and Wholes: Practice (continued)

Part 5 Less than, equal to, or more than 1? Circle one.

29) $\frac{5}{4}$ less than 1 = 1 **(more than 1)**

30) $\frac{3}{8}$ **(less than 1)** = 1 more than 1

31) $\frac{6}{6}$ less than 1 **(= 1)** more than 1

32) $\frac{2}{3}$ **(less than 1)** = 1 more than 1

33) $\frac{9}{8}$ less than 1 = 1 **(more than 1)**

34) $\frac{10}{10}$ less than 1 **(= 1)** more than 1

35) $\frac{7}{5}$ less than 1 = 1 **(more than 1)**

36) $\frac{4}{6}$ **(less than 1)** = 1 more than 1

37) $\frac{3}{3}$ less than 1 **(= 1)** more than 1

38) $\frac{12}{10}$ less than 1 = 1 **(more than 1)**

Fractions and Wholes: Practice (continued)

Part 6 One whole No help this time.

39) $\frac{5}{5} = \boxed{1}$

40) $\frac{\boxed{8}}{8} = 1$

41) $\frac{10}{10} = \boxed{1}$

42) $\frac{\boxed{3}}{3} = 1$

43) $\frac{2}{2} = \boxed{1}$

44) $\frac{\boxed{6}}{6} = 1$

Part 7 Whole numbers No help this time.

45) $\frac{12}{4} = \boxed{3}$

46) $\frac{10}{5} = \boxed{2}$

47) $\frac{6}{2} = \boxed{3}$

48) $\frac{9}{3} = \boxed{3}$

49) $\frac{16}{8} = \boxed{2}$

50) $\frac{12}{6} = \boxed{2}$

Fractions and Wholes: Practice (continued)

Part 7 Whole numbers (continued)

51) $\frac{8}{2} = \boxed{4}$

52) $\frac{15}{5} = \boxed{3}$

53) $\frac{20}{10} = \boxed{2}$

54) $\frac{4}{1} = \boxed{4}$

55) $\frac{18}{6} = \boxed{3}$

56) $\frac{6}{3} = \boxed{2}$

Part 8 Write the whole number as a fraction No help this time.

57) $\frac{\boxed{8}}{2} = 4$

58) $\frac{\boxed{12}}{3} = 4$

59) $\frac{\boxed{16}}{4} = 4$

60) $\frac{\boxed{10}}{2} = 5$

61) $\frac{\boxed{24}}{8} = 3$

62) $\frac{\boxed{30}}{10} = 3$

Fractions and Wholes: Practice (continued)

Part 8 Write the whole number as a fraction (continued)

63) $\frac{6}{1} = 6$

64) $\frac{20}{5} = 4$

65) $\frac{24}{6} = 4$

Part 9 More than one whole No help this time.

66) $\frac{9}{8} = 1 \text{ whole and } \frac{1}{8}$

67) $\frac{7}{5} = 1 \text{ whole and } \frac{2}{5}$

68) $\frac{11}{4} = 2 \text{ wholes and } \frac{3}{4}$

69) $\frac{8}{6} = 1 \text{ whole and } \frac{2}{6}$

70) $\frac{13}{10} = 1 \text{ whole and } \frac{3}{10}$

71) $\frac{5}{2} = 2 \text{ wholes and } \frac{1}{2}$

72) $\frac{10}{3} = 3 \text{ wholes and } \frac{1}{3}$

73) $\frac{11}{6} = 1 \text{ whole and } \frac{5}{6}$

Fractions and Wholes: Practice (continued)

Part 9 More than one whole (continued)

74) $\frac{17}{8} = 2 \text{ wholes and } \frac{\boxed{1}}{8}$

75) $\frac{9}{4} = 2 \text{ wholes and } \frac{\boxed{1}}{4}$

Part 10 Count past 1 No help this time.

76) $\frac{3}{4}\ \frac{4}{4}\ \frac{\boxed{5}}{4}\ \frac{\boxed{6}}{4}\ \frac{7}{4}$

77) $\frac{4}{5}\ \frac{5}{5}\ \frac{\boxed{6}}{5}\ \frac{\boxed{7}}{5}$

78) $\frac{2}{3}\ \frac{3}{3}\ \frac{4}{3}\ \frac{\boxed{5}}{3}\ \frac{\boxed{6}}{3}$

79) $\frac{7}{8}\ \frac{8}{8}\ \frac{\boxed{9}}{8}\ \frac{\boxed{10}}{8}$

80) $\frac{1}{2}\ \frac{2}{2}\ \frac{\boxed{3}}{2}\ \frac{\boxed{4}}{2}\ \frac{5}{2}$

81) $\frac{5}{6}\ \frac{6}{6}\ \frac{7}{6}\ \frac{\boxed{8}}{6}\ \frac{\boxed{9}}{6}$

Part 11 Wholes and pieces

82) $\frac{5}{3} = 1 \text{ whole and } \frac{\boxed{2}}{3}$

83) $\frac{3}{2} = 1 \text{ whole and } \frac{\boxed{1}}{2}$

84) $\frac{23}{12} = 1 \text{ whole and } \frac{\boxed{11}}{12}$

85) $\frac{15}{10} = 1 \text{ whole and } \frac{\boxed{5}}{10}$

Fractions and Wholes: Practice (continued)

Part 11 Wholes and pieces (continued)

86) $\frac{8}{5} = 1 \text{ whole and } \frac{3}{5}$

87) $\frac{4}{3} = 1 \text{ whole and } \frac{1}{3}$

88) $\frac{14}{8} = 1 \text{ whole and } \frac{6}{8}$

89) $\frac{19}{10} = 1 \text{ whole and } \frac{9}{10}$

90) $\frac{17}{10} = 1 \text{ whole and } \frac{7}{10}$

91) $\frac{9}{5} = 1 \text{ whole and } \frac{4}{5}$

92) $\frac{12}{8} = 1 \text{ whole and } \frac{4}{8}$

93) $\frac{6}{4} = 1 \text{ whole and } \frac{2}{4}$

94) $\frac{14}{10} = 1 \text{ whole and } \frac{4}{10}$

95) $\frac{9}{6} = 1 \text{ whole and } \frac{3}{6}$

96) $\frac{10}{8} = 1 \text{ whole and } \frac{2}{8}$

97) $\frac{6}{5} = 1 \text{ whole and } \frac{1}{5}$

Fractions and Wholes: Practice (continued)

Part 11 Wholes and pieces (continued)

98) $\frac{10}{6} = 1 \text{ whole and } \frac{\boxed{4}}{6}$

99) $\frac{16}{12} = 1 \text{ whole and } \frac{\boxed{4}}{12}$

100) $\frac{14}{12} = 1 \text{ whole and } \frac{\boxed{2}}{12}$

101) $\frac{19}{12} = 1 \text{ whole and } \frac{\boxed{7}}{12}$

102) $\frac{22}{12} = 1 \text{ whole and } \frac{\boxed{10}}{12}$

103) $\frac{21}{12} = 1 \text{ whole and } \frac{\boxed{9}}{12}$

104) $\frac{13}{8} = 1 \text{ whole and } \frac{\boxed{5}}{8}$

105) $\frac{7}{6} = 1 \text{ whole and } \frac{\boxed{1}}{6}$

Part 12 Whole numbers as fractions

106) $2 = \frac{\boxed{16}}{8}$

107) $3 = \frac{\boxed{12}}{4}$

108) $2 = \frac{\boxed{8}}{4}$

Fractions and Wholes: Practice (continued)

Part 12 Whole numbers as fractions (continued)

109) $4 = \frac{8}{2}$

110) $5 = \frac{20}{4}$

111) $2 = \frac{4}{2}$

112) $5 = \frac{10}{2}$

113) $4 = \frac{24}{6}$

114) $5 = \frac{30}{6}$

115) $2 = \frac{12}{6}$

116) $3 = \frac{9}{3}$

117) $2 = \frac{6}{3}$

118) $5 = \frac{40}{8}$

119) $3 = \frac{18}{6}$

120) $4 = \frac{16}{4}$

Fractions and Wholes: Practice (continued)

Part 12 Whole numbers as fractions (continued)

121) $5 = \frac{15}{3}$

122) $4 = \frac{32}{8}$

123) $4 = \frac{12}{3}$

Part 13 Remember? Problems from earlier packets.

124) **9 cm by 3 cm: area =** 27 **sq cm**

125) **Round 436 to the nearest ten:** 440

126) **two tenths =** $\frac{2}{10}$

127) **6 groups of 8 =** 48

128) **4 pieces of** $\frac{1}{8} = \frac{4}{8}$

129) **Round 819 to the nearest hundred:** 800

Part 14 Word problems Draw the bars.

130) Ana has 2 pizzas. Each one is cut into 4 equal slices. How many fourths are there in all?

Number sentence **2 wholes = 8/4**

Answer **8** fourths

Draw or show your work

131) Lee needs 1 cup of milk but only has a 1/3 cup scoop. How many scoops make 1 cup?

Number sentence **3/3 = 1**

Answer **3** scoops

Draw or show your work

132) Max eats 6 halves of apples. How many whole apples is that?

Number sentence **6/2 = 3**

Answer **3** apples

Draw or show your work

Fractions and Wholes: Practice (continued)

Part 14 Word problems (continued)

133) Sam has 7 pieces of two same-size bars. Each bar has 4 pieces. Is that more than 1 bar?

Number sentence **7/4 = 1 whole and 3/4**

Answer **yes**

Draw or show your work

134) Challenge: Jo has 10 eighths of a pie. How many whole pies is that, and how many eighths?

Number sentence **10/8 = 1 whole and 2/8**

Answer **1 and 2/8** pies

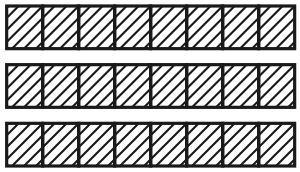
Draw or show your work

Check-In: Fractions and Wholes

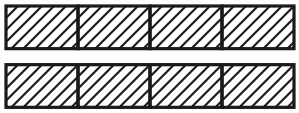
Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 How many wholes?

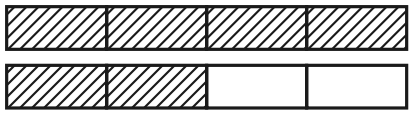
1) 

$\frac{24}{8} = 3$

2) 

$\frac{8}{4} = 2$

Part 2 More than one whole

3) 

$\frac{6}{4}$ = 1 whole and $\frac{2}{4}$

Part 3 Less than, equal to, or more than 1?

4) $\frac{5}{3}$ less than 1 = 1 **more than 1**

5) $\frac{11}{6}$ less than 1 = 1 **more than 1**

Part 4 Wholes

6) $\frac{8}{8} = 1$

7) $\frac{18}{6} = 3$

8) $\frac{30}{10} = 3$

Part 5 Past one whole

9) $\frac{15}{8}$ = 1 whole and $\frac{7}{8}$

10) $\frac{8}{3}$ = 2 wholes and $\frac{2}{3}$